Problem Set 4
ECO3121 - Fall 2022

November 29, 2022

Due 11:59 pm, 12/10/2022

Question 1

A researcher is interested in analyzing the effect of a free fertilizer policy on crop yield. He has a panel data set for 1000 villages in rural China over 2016 to 2019 in which the data for the average crop yield (Y_{it}) in village i in year t and the indicator for whether the village participate in this free fertilizer program (X_{it}) are available.

1. He considers the following panel model:

$$Y_{it} = \beta_1 X_{it} + \alpha_i + u_{it}$$

   where α_i are i.i.d. unobserved random variables and α_i is correlated with X_{it}. Explain how the researcher should estimate β_1 with the data he has and provide a factor that is modelled by α_i. (1 point)

2. He considers the following panel model:

$$Y_{it} = \beta_1 X_{it} + \gamma_t + u_{it}$$

   where γ_t are i.i.d. unobserved random variables and γ_t is correlated with X_{it}. Explain how the researcher should estimate β_1 with the data he has and provide a factor that is modelled by γ_t. (1 point)

3. He considers the following panel model:

$$Y_{it} = \beta_1 X_{it} + \alpha_i + \gamma_t + u_{it}$$

   What is the advantage of this model compared to the above two? Explain the "demean" procedure in estimating the model. (2 point)

Question 2

Following a national poverty alleviation program in Gansu province, in July 2005, all farmers in Gansu province received free fertilizer. Thus, this policy experiment induced a geographical allocation of fertilizers that can be presumed exogenous in a crop yield

regression. Let Y_i^0 and Y_i^1 denote the crop in location i before and after the policy intervention. Let D_i be a dummy variable that takes the value 1 if land parcel i is in Gansu province, 0 otherwise. We would like to estimate the treatment effect of the fertilizer usage on crop yield with the difference-in-difference estimator.

1. (2 points) Write down this difference-in-difference estimator as a function of $\{Y_i^0, Y_i^1, D_i\}$ for $i = 1, ..., n$.

2. (1 points) What is the key assumption for this estimator to be an unbiased estimator of the treatment effect?

3. (2 points) Suppose after the policy intervention, farmers in Gansu province works harder and thus higher crop yield is expected. Does the difference-in-difference estimator under or over estimate the treatment effect. Explain.

Question 3

We are going to replicate a study conducted by Card and Krueger in 1994 that investigate the relationship between a rise in minimum wage and employment.

Economic theories have long suggested that increases in the minimum wage lead to a reduction in the employment for at least two reasons: Businesses are less likely to hire and will rather invest in other resources that are now cheaper because of wage increase. Higher salaries will induce businesses to raise their prices to compensate their greater costs; as prices increase, we expect fewer buyers, which will lead to lower demand and employment.

These theories have found mixed support but the discussion is still very much open within the policy world, as states discuss the opportunity to rise their minimum wage to help local populations to face increasing living costs. Discussions are currently occuring in New Jersey and Illinois to raise the minimum wage to 15$/hour (New york has successfully passed this same raise in 2018).

One of the first study looking at this policy problem was Card and Krueger's. They applied a difference-in-difference the design to look at two groups of fast-food restaurants: fast-food restaurants in New Jersey where the minimum wage increased from 4.25$ to 5.05$ per hour (treatment group) AND fast-food restaurants in Pennsylvania where the minimum wage did not increase (control group). They collected employment data before and after the minimum wage was approved.

1. Explain why comparing New Jersey vs. PA after the change in the minimum wage may not be a good estimation for the treatment effect. (1 point)

2. What is the regression model you'd like to estimate? (1 point)

3. What is the key assumption of the estimation method you use? (1 point)

4. The data is in the blackboard (bb) Assignment 4 file folder titled *DID_Example.csv*, and the variables are defined as in Table 1. Run the regression model you proposed in 2. and report the regression results. (Please include control variables that you think necessary) (2 point)

Table 1: Variable Description

Variable name	Description
ID	Unique identifier for fast food
Treatment	Pre-treatment (=0) and post-treatment (=1)
Group	1 if NJ (treatment); 0 if PA (Control)
Empl	# of full time employees
C.Owned	If owned by a company (=1) or not (=0)
Hours.Opening	Number hours open per day
Soda	Price of medium soda, including tax
Fries	price of small fries, including tax
Chain	1 = BK, 2 = KFC, 3 = Roys, 4 = Wendys
SouthJ	South New Jersey
CentralJ	Central New Jersey
NorthJ	North New Jersey
PA1	Northeast suburbs of Philadelphia
PA2	Easton and other PA areas
Shore	New Jersey Shore

5. Explain why this is an unbiased estimation of the treatment effect. (Hint: You can use a graph if necessary) (1 point)